GG-S-805B
November 18, 1977
SUPERSEDING
Int. Fed. Spec. GG-S-00805a (DSA-DM)
February 8, 1967 and
Fed. Spec. GG-S-805
June 26, 1950

FEDERAL SPECIFICATION

SURVEYOR, DENTAL CLASP

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a clasp surveyor suitable for use in designing dental prosthesis.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 151 - Metal, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

FSC 6520

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Forth Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation-packaging, Methods Of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Drawing:

Defense Personnel Support Center (DPSC)

24048 - Surveyor, Dental Clasp

(Copies of Military Specifications, Standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The dental surveyor's components shall be fabricated of the materials shown in table I. Materials shall conform to the applicable specification and requirements hereinafter specified. Material not specifically designated shall be of high commercial quality throughout and shall be entirely suitable for the purpose intended.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

TABLE I. Materials.

Material	Surveyor components
Cast aluminum	Base; tilt-top table (top, mount, clamp); upright column; surveying arm; tilt-top table locking screw knob.
Nickel-plated brass	Upright column locking screw; surveying spindle locking screw; surveying spindle cap; 2 degree and 6 degree taper tools; tilt-top table fixed jaws (2).
Stainless steel (AISI 416)	Surveying spindle; tilt-top table locking screw; coil spring.
Steel	Upright column stud, washer, nut; tilt-top table ball and socket joint bushing; spindle rotational stop bushing.
Nickel-plated steel	Surveying rod and cylindrical blocking-out tools; under cut gauge; tilt-top table clamp transfer screw.
Tempered steel	Tilt-top table mount ball bearing; spindle rotational stop pin.
Heat treated steel	Spindle collet and three-jawed chuck.
Brass	Tilt-top table locking screw liner; tilt-top table ball bearing bushing liner.
Self-lubricated bronze	Spindle bearings.
Carbon	Spindle scribing rods.
Rubber	Surveyor base feet.
Plastic	Accessory tool and carbon rod containers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

3.2 Design and construction. The dental surveyor shall conform to Defense Personnel Support Center Drawing 24048 for style, design and dimensions. The surveying spindle shall be machined and centerless ground to produce a straight, concentric rod. The surveying spindle shall be mounted on self-lubricating bronze sleeve bearings. A slotted collar shall be installed under the lower bronze bearing and a drive pin inserted through the spindle, which permits removing and adding tools to the chuck with one hand. The lower end of the spindle shall be drilled internally and threaded externally to receive the threaded collet. The collet shall activate the three-jawed chuck used to hold the various interchangeable working tools. The upper end of the upright column and the lower end of the cylinder housing of the surveying arm shall be machined to matching taper bearing surfaces.

3.2.1 Drawings. The drawings forming a part of this specification are not production drawings. Drawings establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the drawings to provide a product with proper design, configuration, fit and function. No deviations from the dimensional limitations prescribed on the drawings are permissible without prior approval of the contracting officer.

3.3 Runout. The surveying spindle shall be perpendicular to the top surface of the surveyor base and shall have a runout not exceeding 0.005 inch when tested as specified in 4.3.1.

3.4 Finish. All surfaces, except for plated hardware components and those listed herein, shall have a black wrinkle, baked enamel finish. The top surface of the surveyor base shall be machined. The bottom, outer rim surface of the tilt-top table mount shall be machined to provide easy movement over the top surface of the surveyor base. The top surface of the tilt-top table shall be smooth, with a matte or anodized finish. The matching taper, bearing surfaces of the surveying arm and upright column shall have a machined finish.

3.5 Instructions. Adequate and accurate operating instructions shall accompany each clasp surveyor.

3.6 Marking. Each clasp surveyor shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.7 Workmanship. The workmanship shall be first class in every respect. The clasp surveyor shall be free from imperfections which affect its appearance or which may impair its serviceability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the metal, rubber, plastic and packaging materials, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component or material, the date of testing, the test method, individual test results, the specification requirements, the lot number, the lot size and the sample size.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties). Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. Surveyors shall be examined for defects, including, but not limited to, those defects listed in table II.

TABLE II. Classification of Defects.

Categories	Defects
Critical	None defined.
Major:	
101	Top surface of base or bottom surface of tilt-top table mount not smooth or level.
102	Base, surveying arm, upright column or tilt-top table has significant casting defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of defects. (Continued)

Categories	Defects
Major:	
103	Burrs, nicks or feathers present on surveyor components.
104	Threads too rough for collet-chuck operation.
105	Exposed surfaces not free from rough edges.
106	Spindle not centerless ground.
107	Surveyor components not properly assembled.
Minor:	
201	Component surfaces not clean or showing corrosion.
202	Plating of components shows signs of peeling or flaking.
203	Enamel finish not as specified.
204	Instrument identification missing.
205	Instrument identification not as specified.

The inspection levels and acceptable quality levels (AQL's) shall be as listed in table III. The unit of product for examination shall be one surveyor.

TABLE III. Examination criteria.

Type defect	Inspection level	AQL (percent defective)
Major	II	2.5
Minor	II	4.0

4.2.2 Testing. Surveyors shall be tested as specified in table IV and 4.3. Inspection levels and acceptable quality levels (AQL's) shall be as listed in table IV. The unit of product for testing shall be one surveyor.

TABLE IV. Test criteria.

Characteristic	Component	Requirement	Test procedure	Inspection level	AQL
Runout	Surveyor	3.3	4.3.1	S-3	1.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

4.2.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE V. Examination of preparation for delivery.

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.2.3.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.3 Tests. Tests shall be conducted to determine compliance with the requirements of this specification.

4.3.1 Runout. The total indicated runout of the surveying spindle shall be measured to the nearest 0.001 inch. The spindle shall be held in its housing (positioned as shown on DPSC Drawing 24048, sheet 1 of 2) and rotated slowly. A dial indicator sensitive to 0.0005 inch, equipped with a knife-edge foot or other suitable equipment of equal accuracy shall be employed. If a dial indicator is used, it shall be mounted with the shaft perpendicular to the axis of the spindle and shall be so positioned that the foot contacts the spindle surface. The difference between the maximum and minimum readings for one full rotation of the spindle shall be taken as the total indicated runout.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

5.1.1 Level A. Each surveyor shall be disassembled as far as is practicable. The components comprising the surveyor shall be wrapped in neutral tissue or other suitable material and packaged in accordance with MIL-P-116, method IC-2 except that a preservative shall not be utilized. As an alternate, each unit may be shrink-wrapped or blister-packed with 0.010 inch clear plastic to a sheet of corrugated fiberboard. The vacuum packaged sheet shall be folded into a box which enclosed the article being packed. Closure shall be effected by use of mylar tape at all seams.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Six unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Six unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies.

5.3.2.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:53Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The dental clasp surveyor covered by this specification is intended for use in designing and constructing removable partial appliances.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Required marking (see 5.3).

6.3 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-558-2700	SURVEYOR, DENTAL CLASP, Stationary.

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DOD - MB

Preparing activity:
DLA-DM

Civil Agency Coordinating Activities:
COMMERCE-NBS
GSA-FSS
HEW-FDA
VA-DMS

Project No. 6520-1408

Orders for this publication are to be placed with General Services Administration, acting as agent for the Superintendent of Documents. See Section 2 of this specification to obtain copies of documents referenced herein. Price 70 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

GG-S-805B SURVEYOR, DENTAL CLASP

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, PA 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.